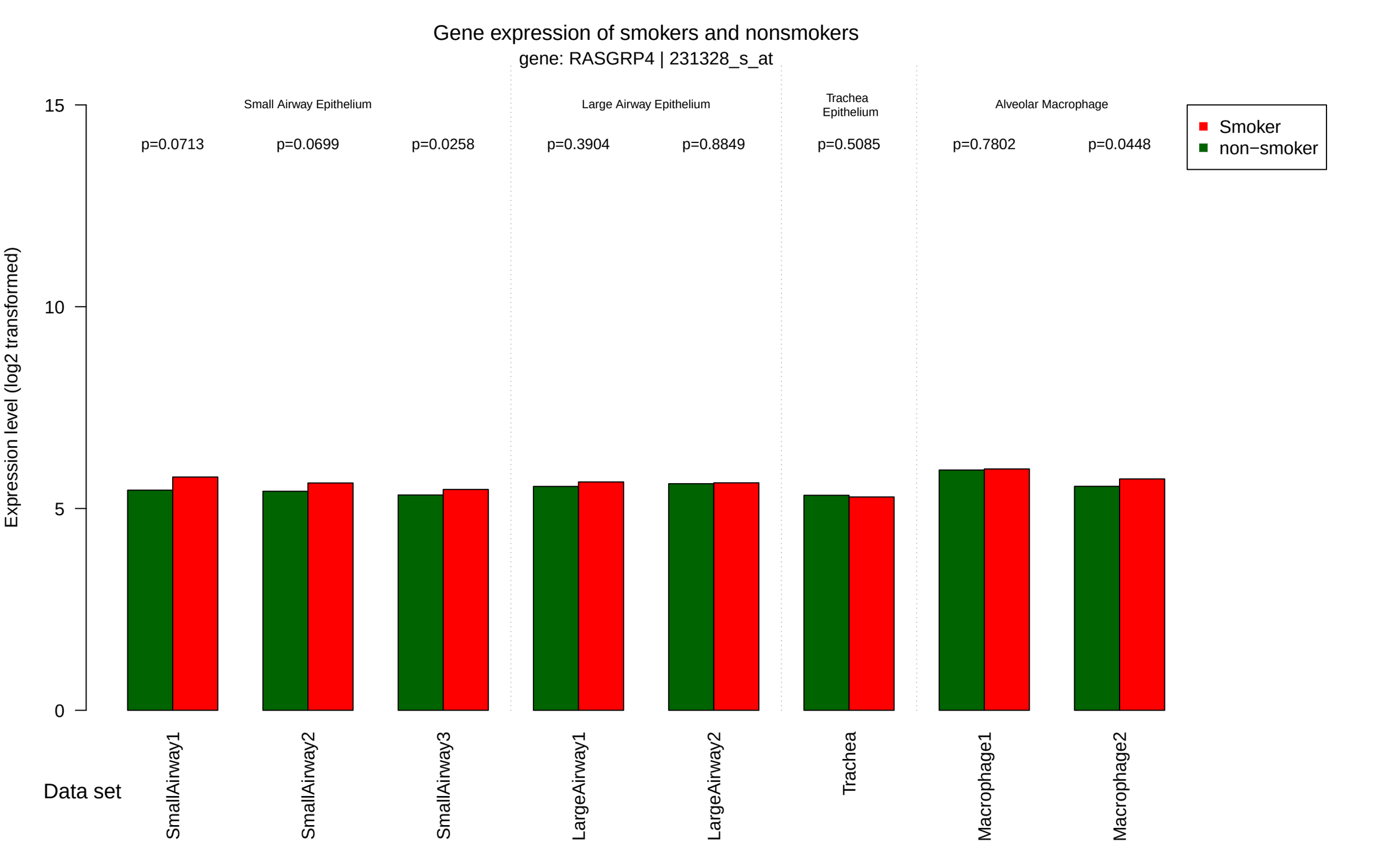
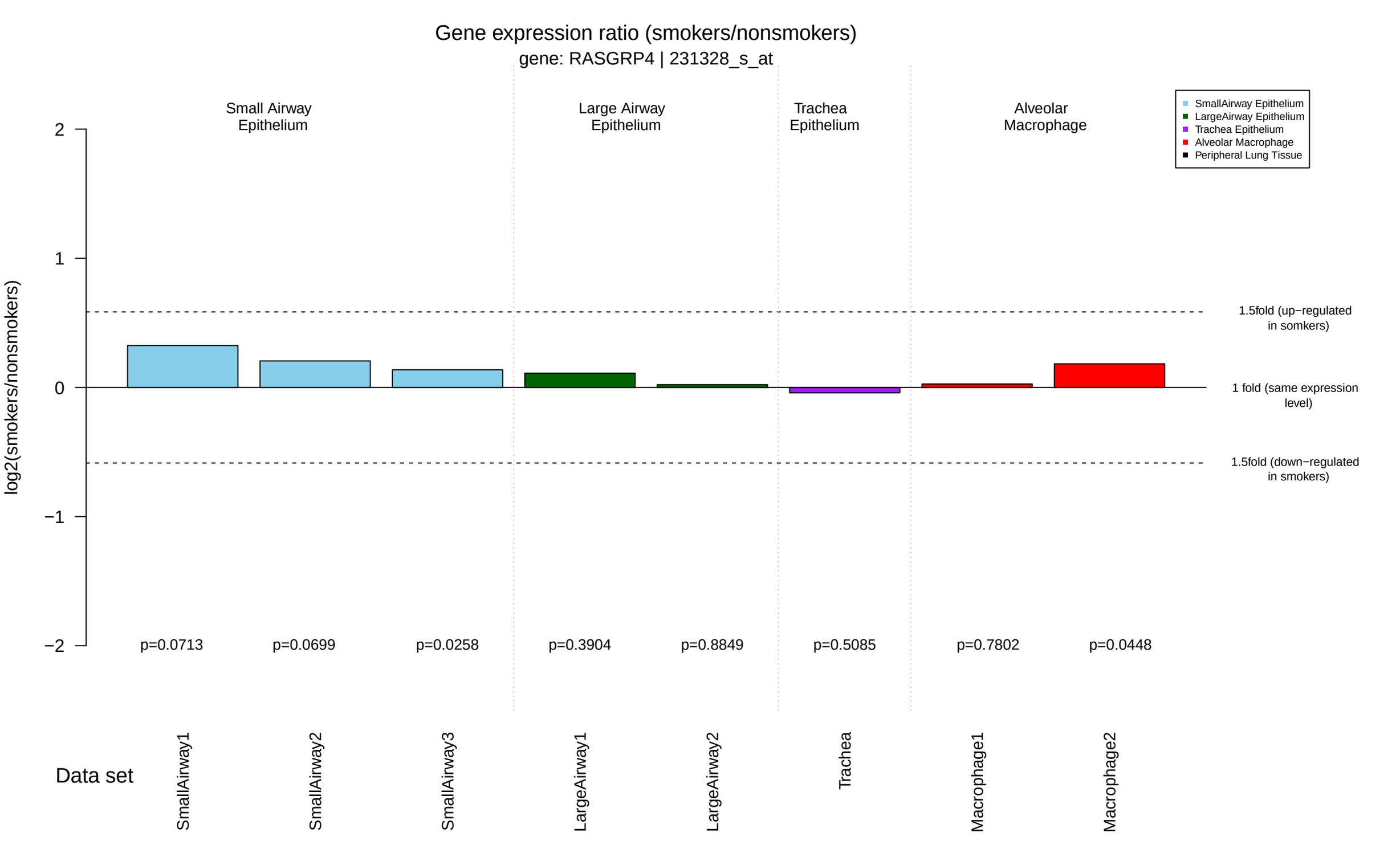
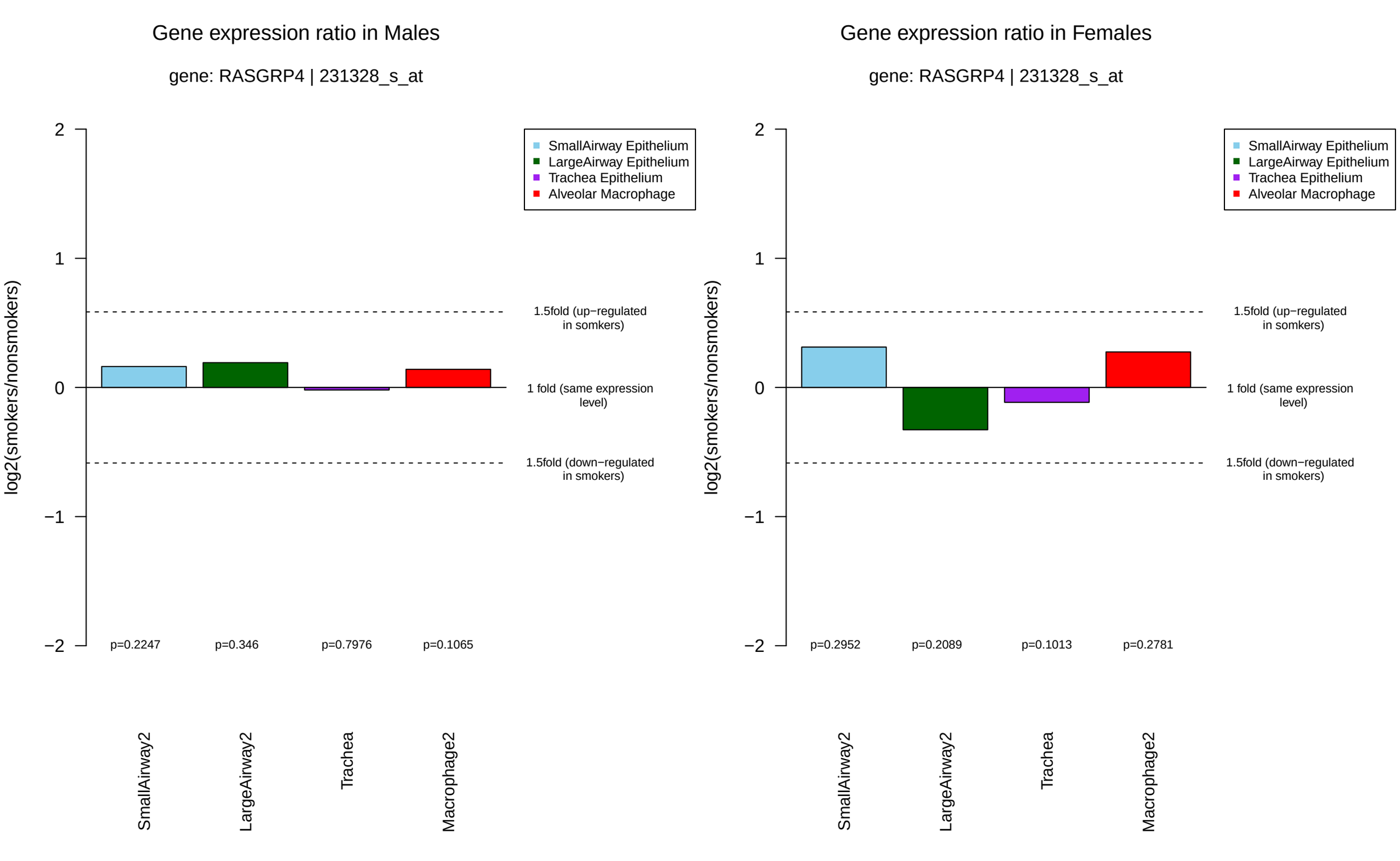
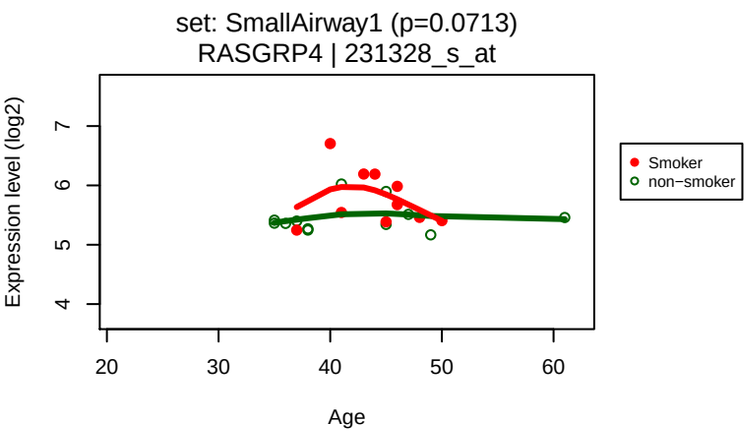
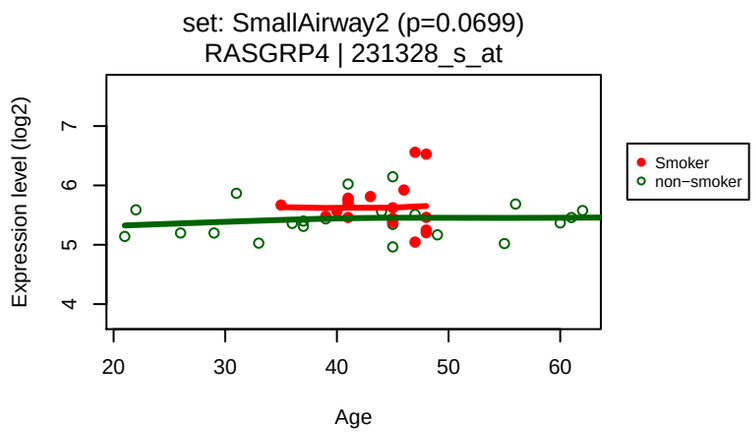
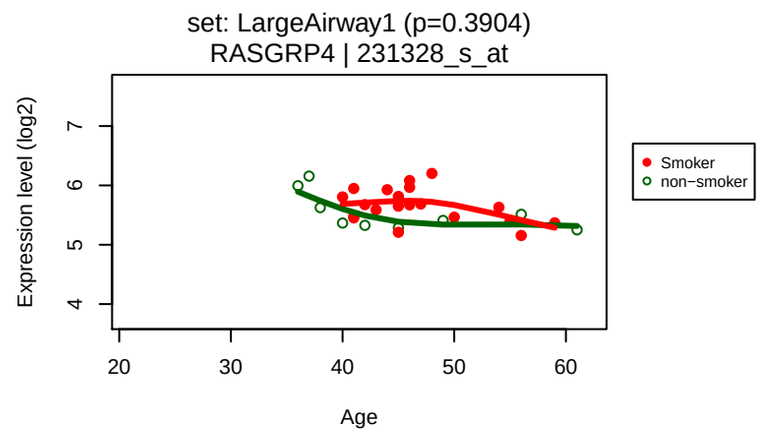

Gene expression of smokers and nonsmokers
gene: RASGRP4 | 231328_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0713
p=0.0699
p=0.0258
p=0.3904
p=0.8849
p=0.5085
p=0.7802
p=0.0448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

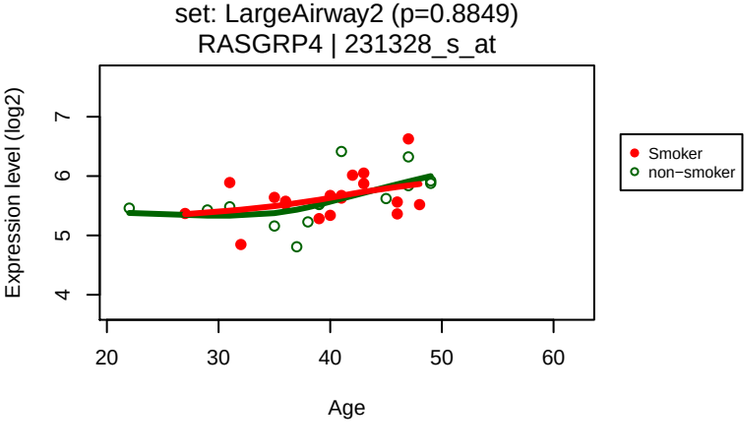

Gene expression ratio (smokers/nonsmokers)
gene: RASGRP4 | 231328_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0713
p=0.0699
p=0.0258
p=0.3904
p=0.8849
p=0.5085
p=0.7802
p=0.0448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

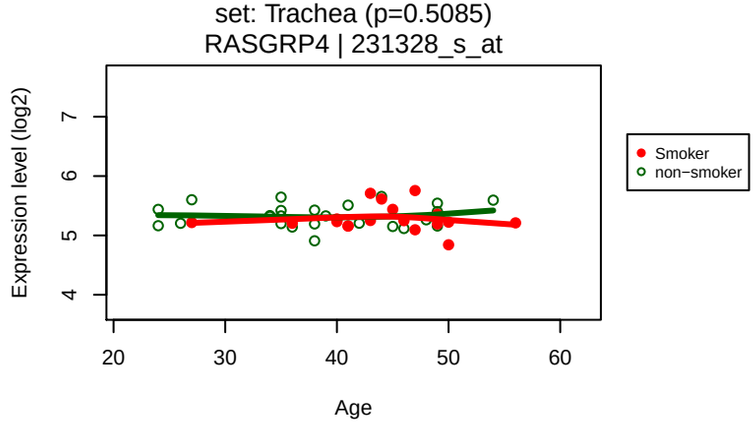

Gene expression ratio in Males
gene: RASGRP4 | 231328_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2247
p=0.346
p=0.7976
p=0.1065
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RASGRP4 | 231328_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2952
p=0.2089
p=0.1013
p=0.2781
SmallAirway2
LargeAirway2
Trachea
Macrophage2

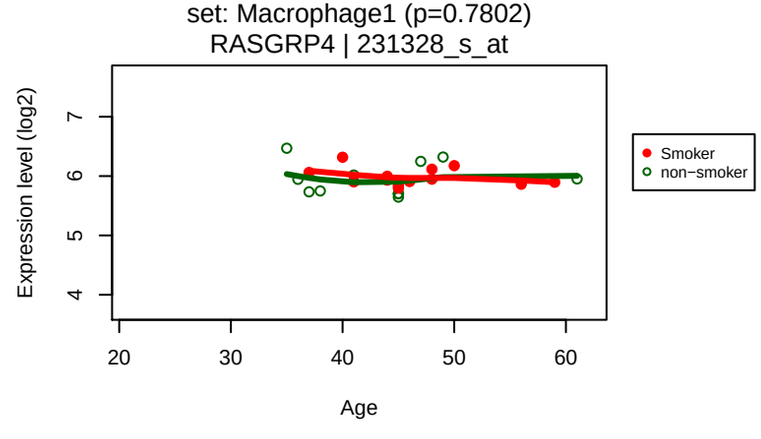

set: SmallAirway1 (p=0.0713)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker

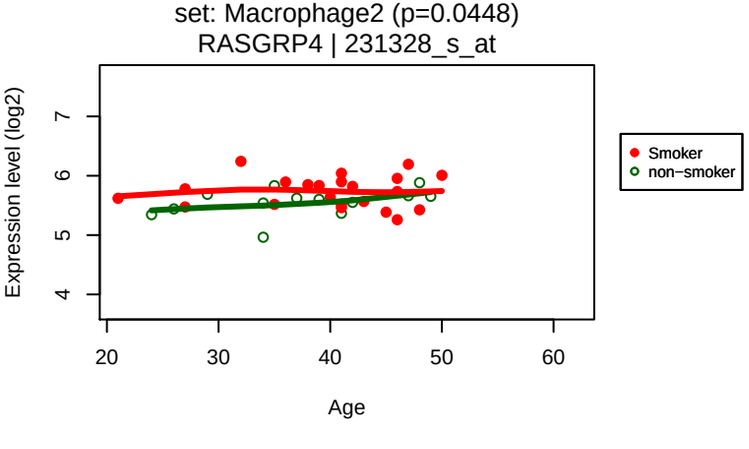

set: SmallAirway2 (p=0.0699)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker

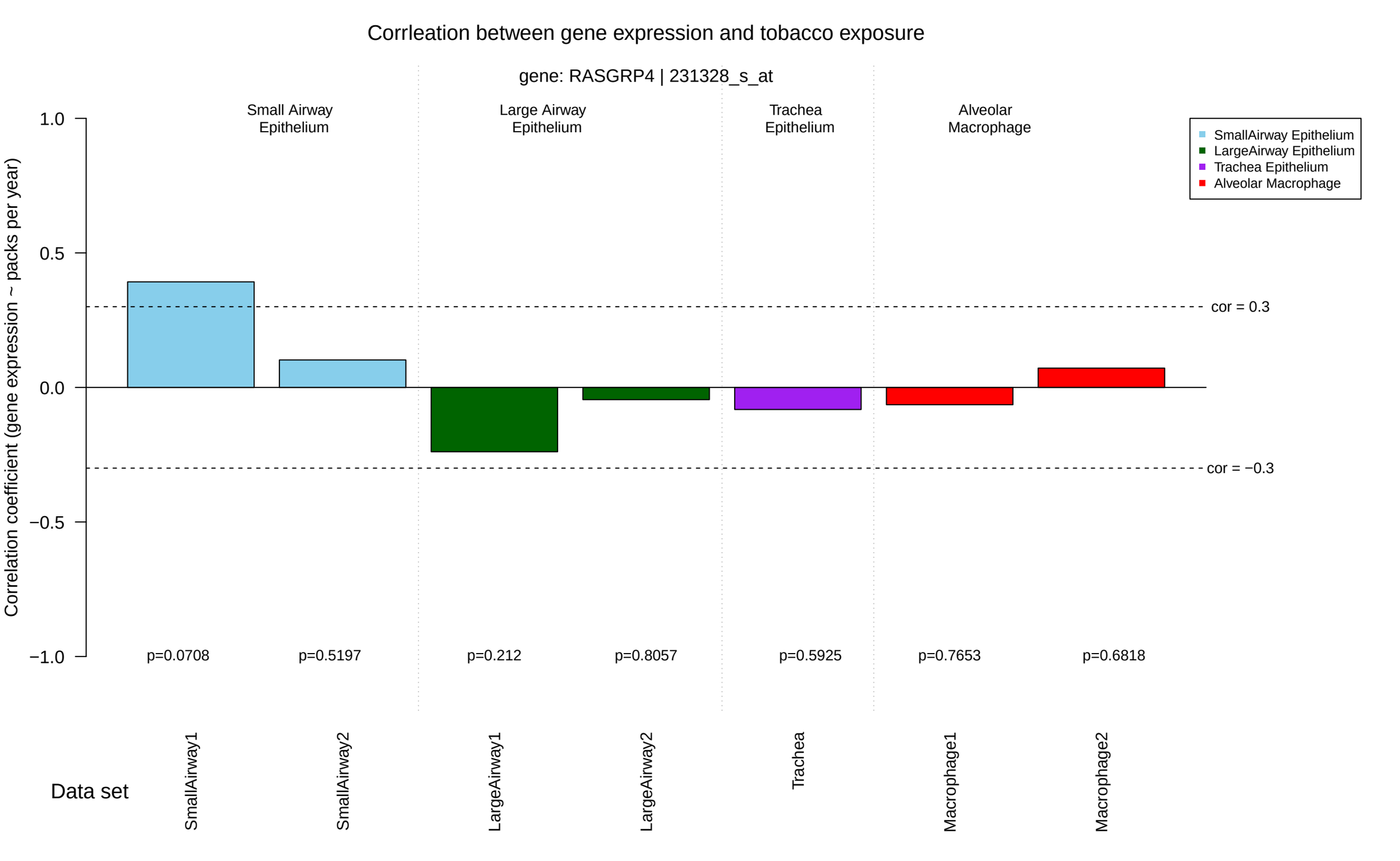

set: LargeAirway1 (p=0.3904)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8849)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5085)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7802)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0448)
RASGRP4 | 231328_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RASGRP4 | 231328_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0708
p=0.5197
p=0.212
p=0.8057
p=0.5925
p=0.7653
p=0.6818
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2